Five-Year Trends in Reported National Football League Injuries

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Abstract

Objective: To determine the incidence of all-cause injury and concussion in the National Football League (NFL) over a 5-season time span (2012-2016). **Design:** Prospective descriptive epidemiological study. **Setting:** National Football League Injury Report data from 2012 to 2016. **Participants:** National Football League players. **Interventions:** None (descriptive study). **Main Outcome Measures:** Injury report data were collected prospectively for all NFL injuries from 5 seasons (2012-2016). The incidences of reported concussions, knee injuries, and all-cause injury were compared across the 5 seasons using the Kruskal–Wallis rank-sum test. **Results:** A total of 10 927 injuries were identified across the 5 seasons, including 752 (6.9%) concussions. The top 3 most injured areas included the knee (17.2%), ankle (13.6%), and shoulder (8.8%). Defensive backs consistently had the highest number of all-cause injuries per season. When comparing across years, there was a significant decrease in all-cause injury in 2016 compared with 2015, a significant decrease in knee injuries in 2016 compared with 2015, and a significant increase in concussion in 2015 compared with 2014 ($P < 0.05$). **Conclusions:** Reported all-cause injury incidence and knee injury incidence is currently on the decline. However, reported concussion incidence has recently increased, perhaps due to increased awareness and rule changes implemented to aid in the detection and treatment of concussion. Strategies to reduce injury and improve injury awareness should continue to be explored.

Key Words: National Football League, injury, concussion, epidemiology

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BACKGROUND

The risk of injury associated with football is significant with all-cause injury incidence rates reported to be as high as 20.9 per 1000 athletes at risk (AAR) in high school football,[1] 44.2 per 1000 AAR in college football,[2] and 395.8 per 1000 AAR in the National Football League (NFL).[3] Moreover, football has one of the highest estimates of traumatic brain injury incidence of all the major sports.[4] Research into the long-term impact of multiple concussions and repetitive traumatic brain injuries is evolving with evidence identifying a potential negative effect on long-term neurocognitive outcomes.[1] The high incidence of concussion and all-cause injury in football is of concern, considering the large number of participants and potential for hazardous long-term outcomes.[5,6] Establishing injury rates in football is important to assess current- and inform future-injury prevention initiatives, monitor injury trends, and better inform participants of the potential risks of participation.

The NFL's influence on the rules, policies, and injury prevention strategies at all levels of play is undeniable.[7] The NFL has been striving to better understand injury causes and develop strategies to reduce injury rates through rule changes and educational campaigns.[8–12] The NFL annually reviews it policies and rules, and implements rule changes in an aim to increase player safety. Evaluating the effectiveness of these rule changes is critical to evaluate their utility and inform further injury prevention initiatives. Establishing injury rates is a fundamental component to determining the effectiveness of injury prevention strategies, including rule and policy changes.

In a previous pilot study examining NFL injury report data over 2 seasons from 2012 to 2014, an all-cause injury rate of 395.8 per 1000 AAR and concussion incidence of 27.8 per 1000 AAR were identified.[3] In comparing the incidence of concussion in the 2012 to 2014 seasons with the seasons of 2002 to 2007,[11] it was found that the incidence was 1.61-fold higher in the more recent time period.[3] Expanding the independent peer-reviewed surveillance of NFL injuries is warranted to evaluate attempts to reduce injuries, in addition to provide an accurate estimate of the risk associated with the sport at this level of play.

The purpose of this study was to document the incidence of all-cause, musculoskeletal, and concussion injury rates in the NFL over 5 seasons from 2012 to 2017, with consideration of the change in rate of all-cause, knee, and concussion injuries.

METHODS

National Football League Injury Reports

Injury report data were collected from Official NFL injury reports. These reports are published online weekly by each of the 32 teams in the league on each team's web site and on the NFL league web site.[13] Each team is required by the Personal (Injury) Report policy to "provide a full and complete rendering

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TABLE 1. Team Games Included

	TGs, n (%)
Regular season (16 games, 17 weeks)	2560 (100.0)
Weeks 1 through 16 (games 1-15)	2400 (93.8)
Week 17 (game 16) with available injury report	60 (2.3)
Included in this study	2460 (96.1)

of player availability. […] The information must be credible, accurate, and specific […] A violation of the policy may result in Commissioner discipline, which may include a fine on the involved club, fines or suspensions of involved individuals, as well as the possible forfeiture of draft choices by the involved club."[14]

National Football League injury reports are published before each game and, therefore, reflect injuries sustained during the previous week's game or in the current week's practice. Teams play 16 regular-season games in a 17-week period. Only the 12 teams that advance to the playoffs are required to publish a report to reflect the last regular-season game. Therefore, injury reports for the last regular-season game are not available for the 20 teams that are eliminated. In addition to the injury reports, injured players who are not immediately available to participate may be placed on an injured reserve list and not the weekly injury report. This list is available online and is updated daily.

Injury reports and the injured reserve list include the following information of all players on the list who have been diagnosed by a team physician: player name, position, anatomical area injured, practice status of the player for the practices during the week (did not practice, limited participation, and full participation), and game status (probable to play, doubtful to play, and out).

Data Collection and Injury Definition

All injury reports and injury reserve data were collected prospectively for each week over the 2012 to 2013, 2013 to 2014, 2014 to 2015, 2015 to 2016, and 2016 to 2017 regular seasons for all 32 teams. Week 17 injury reports were only available for the 12 teams that progressed to the playoffs. An injury was defined as any reported injury not documented in the previous week's injury report. Injury reports do not document recurring injuries; however, using this definition ensures that the injury was not affecting the player in the preceding week before its inclusion. A concussion was defined as any injury reported as a concussion or head injury.

Data Analysis and Ethics

Injury reports do not indicate whether an injury was sustained during a game or a practice; however, previous suggestions indicate that most injuries occur during games.[15,16] Therefore,

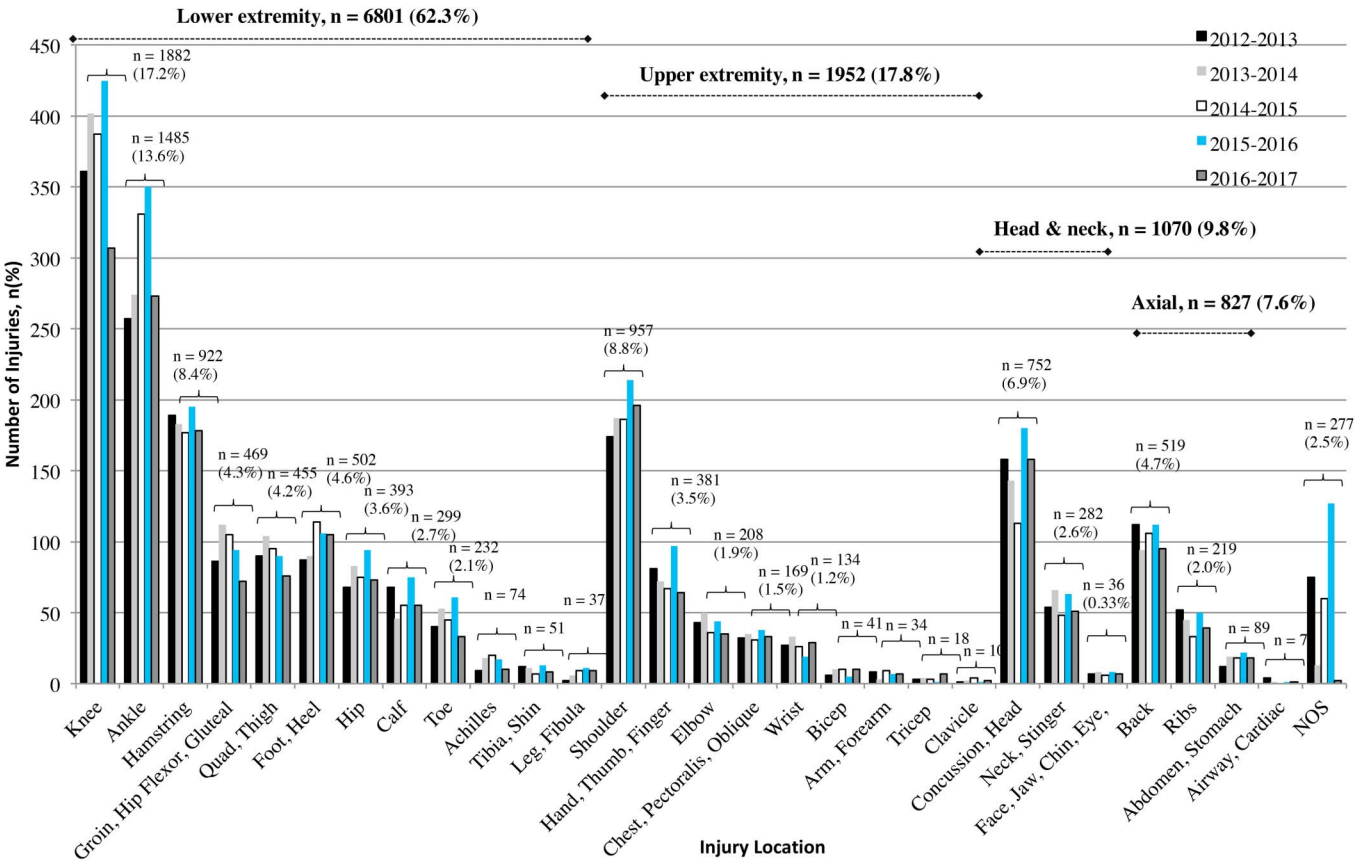

Figure 1. Reported injuries (n, %) by anatomic location for each season (2012 to 2016). NOS, not otherwise specified.

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TABLE 2. Incidence of All-Cause Injury by Position per Year

Position	2012-2013 Injuries (n, %)	2013-2014 Injuries (n, %)	2014-2015 Injuries (n, %)	2015-2016 Injuries (n, %)	2016-2017 Injuries (n, %)	2012-2017 Total Injuries (n, %)
Wide receiver	315 (14.9)	281 (13.0)	265 (12.2)	337 (13.4)	254 (13.0)	1452 (13.3)
Tight end	131 (6.2)	139 (6.4)	151 (6.9)	181 (7.2)	142 (7.3)	744 (6.8)
Defensive back	454 (21.4)	475 (21.9)	508 (23.3)	573 (22.7)	416 (21.3)	2426 (22.2)
Running back	203 (9.6)	229 (10.6)	206 (9.5)	221 (8.8)	152 (7.8)	1011 (9.2)
Linebacker	301 (14.2)	333 (15.4)	355 (16.3)	368 (14.6)	302 (15.5)	1659 (15.2)
Defensive line	321 (15.2)	304 (14.0)	312 (14.3)	339 (13.5)	277 (14.2)	1553 (14.2)
Offensive line	312 (14.7)	320 (14.8)	306 (14.1)	371 (14.7)	327 (16.8)	1636 (15.0)
Quarterback	54 (2.5)	65 (3.0)	56 (2.6)	100 (4.0)	67 (3.4)	342 (3.1)
Kicker, punter	27 (1.3)	21 (1.0)	18 (0.8)	30 (1.2)	15 (0.8)	111 (1.0)
Total	2118 (100.0)	2167 (100.0)	2177 (100.0)	2520 (100.0)	1952 (100.0)	10 934 (100.0)

Position	2012-2013 Injuries per 100 TGs (95% CI)	2012-2013 Injuries per 1000 AAR (95% CI)	2013-2014 Injuries per 100 TGs (95% CI)	2013-2014 Injuries per 1000 AAR (95% CI)	2014-2015 Injuries per 100 TGs (95% CI)	2014-2015 Injuries per 1000 AAR (95% CI)
Wide receiver	64.02 (56.95-71.09)	58.20 (51.78-64.63)	57.11 (50.44-63.79)	51.92 (45.85-57.99)	53.86 (47.38-60.35)	48.97 (43.07-54.86)
Tight end	26.63 (22.07-31.19)	24.21 (20.06-28.35)	28.25 (23.56-32.95)	25.68 (21.41-29.95)	30.69 (25.80-35.59)	27.90 (23.45-32.35)
Defensive back	92.28 (83.79-100.76)	83.89 (76.17-91.60)	96.54 (87.86-105.23)	87.77 (79.87-95.66)	103.25 (94.27-112.23)	93.87 (85.70-102.03)
Running back	41.26 (35.58-46.94)	37.51 (32.35-42.67)	46.54 (40.52-52.57)	42.31 (36.83-47.79)	41.87 (36.15-47.59)	38.06 (32.87-43.26)
Linebacker	61.18 (54.27-68.09)	55.62 (49.33-61.90)	67.69 (60.41-74.95)	61.53 (54.92-68.14)	72.15 (64.65-79.66)	65.59 (58.77-72.42)
Defensive line	65.24 (58.11-72.38)	59.31 (52.82-65.80)	61.79 (54.84-68.73)	56.17 (49.86-62.49)	63.41 (56.38-70.45)	57.65 (51.25-64.05)
Offensive line	63.41 (56.38-70.45)	57.65 (51.25-64.05)	65.04 (57.91-72.17)	59.13 (52.65-65.61)	62.20 (55.23-69.16)	56.54 (50.21-62.88)
Quarterback	10.98 (8.05-13.9)	9.98 (7.32-12.64)	13.21 (10.00-16.42)	12.01 (9.09-14.93)	11.38 (8.40-14.36)	10.35 (7.64-13.06)
Kicker, punter	5.49 (7.56-3.42)	4.99 (3.11-6.87)	4.27 (2.44-6.09)	4.99 (3.11-6.87)	3.66 (1.97-5.35)	3.33 (1.79-4.86)
Total	430.49 (412.15-448.82)	391.35 (374.69-407.02)	440.45 (421.90-458.99)	400.41 (383.55-417.27)	442.48 (423.89-461.07)	402.25 (385.36-419.15)

Position	2015-2016 Injuries per 100 TGs (95% CI)	2015-2016 Injuries per 1000 AAR (95% CI)	2016-2017 Injuries per 100 TGs (95% CI)	2016-2017 Injuries per 1000 AAR (95% CI)	2012-2017 Total Injuries per 100 TGs (95% CI)	2012-2017 Total Injuries per 1000 AAR (95% CI)
Wide receiver	68.50 (61.18-75.81)	62.27 (55.62-68.92)	51.63 (45.28-57.98)	46.93 (41.16-52.70)	59.02 (55.99-62.06)	53.66 (50.90-56.42)
Tight end	36.79 (31.43-42.15)	33.44 (28.57-38.32)	28.86 (24.11-33.61)	26.24 (21.92-30.55)	30.24 (28.07-32.42)	27.49 (25.52-29.47)
Defensive back	116.46 (106.93-126.00)	105.88 (97.21-114.54)	84.55 (76.43-92.68)	76.87 (69.48-84.25)	98.62 (94.69-102.54)	89.65 (86.09-93.22)
Running back	44.92 (39.00-50.84)	40.84 (35.45-46.22)	30.89 (25.98-35.81)	28.09 (23.62-32.55)	41.10 (38.56-43.63)	37.36 (35.06-39.66)
Linebacker	74.80 (67.15-82.44)	68.00 (61.05-74.94)	61.38 (54.46-68.31)	55.80 (49.51-62.10)	67.44 (64.19-70.68)	61.31 (58.36-64.26)
Defensive line	68.90 (61.57-76.24)	62.64 (55.97-69.31)	56.30 (49.67-62.93)	51.18 (45.16-57.21)	63.13 (59.99-66.27)	57.39 (54.54-60.25)
Offensive line	75.41 (67.73-83.08)	68.55 (61.58-75.53)	66.46 (59.26-73.67)	60.42 (53.87-66.97)	66.50 (63.28-69.73)	60.46 (57.53-63.39)
Quarterback	20.33 (16.34-24.31)	18.48 (14.86-22.10)	13.62 (10.36-16.88)	12.38 (9.42-15.34)	13.90 (12.43-15.38)	12.94 (11.30-13.98)
Kicker, punter	6.10 (3.92-8.28)	5.54 (3.56-7.53)	3.05 (1.51-4.59)	2.77 (1.37-4.17)	4.51 (3.67-5.35)	4.10 (3.34-4.87)
Total	512.20 (492.20-532.19)	465.63 (447.45-483.81)	396.75 (379.15-414.35)	360.68 (344.68-376.68)	444.47 (436.14-452.80)	404.07 (396.49-411.64)

all injuries were assumed to have occurred during games for injury rate calculations.

Injuries were reported as the number of injuries and injuries per team game (TG). A TG is defined as a team exposure to a game, and for each game played, there are 2 TGs. A total of 512 TGs are played each season; however, the 20 teams that do not make playoffs are not required to publish an injury report for their final game. Therefore, only 492 TGs per season were available for this study. Descriptive statistics were reported as total injuries (n and %). To determine differences in injury rates between the years, the number of injuries per TG was calculated by dividing the total number of injuries by the number of TGs for each week of play.

To estimate the overall injury burden, an AAR methodology was used.[3,11] At any given time, there are 11 AARs per TG (ie, 11 athletes on the field per team at any one time exposed to risk). Therefore, the denominator for this calculation of AAR injury rate reporting was the number of AARs multiplied by the number of TGs ($11 \times 2460 = 27\,060$).

Given the nonparametric nature of the data, a Kruskal–Wallis (KW) rank-sum test was used to compare year-by-year trends of injury rates. Post hoc analysis was conducted using pairwise comparisons using the Tukey and Nemenyi test with Tukey-Dist approximation for independent samples. The rejection level for all analyses was $P = 0.05$. Analysis was conducted in R software (v.1.0.136).

RESULTS

A total of 1280 regular-season games or 2560 TGs were played over the 5 seasons, of which 2460 TGs (96.1%) were available and included in this study (Table 1). A total of 10 934 injuries were identified, including 752 (6.9%) concussions. A breakdown of injuries by anatomical location

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TABLE 3. Injuries per TG by Year

Season	Total Injuries	Injuries per TG
2012-2013	2118	4.3
2013-2014	2167	4.4
2014-2015	2177	4.4
2015-2016	2520	5.1
2016-2017	1952	4.0

revealed that 6801 (62.3%) were to the lower extremities, 1952 (17.8%) to the upper extremities, and 1070 (9.8%) to the head and neck (Figure 1). The top 5 most injured areas included the knee (n = 1882, 17.2%), ankle (n = 1485, 13.6%), shoulder (n = 957, 8.8%), hamstring (n = 922, 8.4%), and concussion (n = 752, 6.9%). The all-cause injury rate over 5 seasons was 444.47 per 100 TGs [95% confidence interval (CI), 436.14-452.80] or 404.07 per 1000 AAR (95% CI, 396.49-411.64), whereas the all-cause concussion incidence rate was 30.57 per 100 TGs (95% CI, 28.38-32.75) or 27.79 per 1000 AAR (95% CI, 25.80-29.78).

Incidence breakdown by the position indicated that defensive backs consistently had the highest number of all-cause injuries per season, followed by linebackers, defensive line, and offensive line, whereas kickers/punters had the lowest all-cause injuries per season (Table 2).

Comparison of year-by-year trends yielded statistically significant differences in incidence across the years. The incidence of all-cause injuries was significantly lower in 2016 [396.75 per 100 TGs (95% CI, 379.15-414.35) or 360.68 per 1000 AAR (95% CI, 344.68-376.68)] compared with 2015 [512.20 per 100 TGs (95% CI, 492.20-532.19) or 465.63 per 1000 AAR (95% CI, 447.45-483.81); KW; $H(4) = 16.254$, $P = 0.030$, T&M, $P < 0.001$; Tables 3 and 4]. There was no statistically significant difference between the other years. The incidence of knee injuries was significantly less in 2016 [62.40 per 100 TGs (95% CI, 55.42-69.38) or 56.73 per 1000 AAR (95% CI, 50.38-63.07)] compared with 2015 [86.38 per 100 TGs (95% CI, 78.17-94.59) or 78.53 per 1000 AAR (95% CI, 71.06-86.00); KW, $H(4) = 10.709$, $P = 0.030$; T&N, $P = 0.029$; Table 4]. The incidence of concussion was significantly higher in 2015 [36.59 per 100 TGs (95% CI, 31.24-41.93) or 33.26 per 1000 AAR (95% CI, 28.40-38.12)] compared with 2014 [22.97 per 100 TGs (95% CI, 18.73-27.20) or 20.88 per 1000 AAR (95% CI, 17.03-24.73); KW, $H(4) = 10.552$, $P = 0.032$; T&N, $P = 0.012$; Table 4].

DISCUSSION

This study documented the incidence of NFL injuries over 5 seasons from 2012 to 2016 and documented an all-cause injury rate of 444.47 per 100 TGs (95% CI, 436.14-452.80) or 404.07 per 1000 AAR (95% CI, 396.49-411.64) and concussion incidence rate of 30.57 per 100 TGs (95% CI, 28.38-32.75) or 27.79 per 1000 AAR (95% CI, 25.80-29.78). These injury rates are higher than the rates in other collision sports such as ice hockey and rugby.[17,18]

Furthermore, through assessment of year-by-year trends, all-cause injury and knee injury incidence rates decreased in the 2016 in comparison with the 2015 season. The change in all-cause and knee injury rates potentially reflects a protective influence on the rule changes implemented between these 2 seasons including making all chop blocks illegal, expansion of the horse collar rule, and disqualification of a player who is penalized twice in one game for unsportsmanlike conduct.[9] The reduction in knee injury incidence may be attributed specifically to the removal of chop blocks and expansion of the horse collar rule.[9] Moreover, the observed reduction in knee injury incidence in the most recent season may aid in dissipating player fears[19] that additional concussion rule changes will subsequently increase knee and other injuries. In addition to rule changes aimed at reducing concussion incidence, the league is also adjusting other rules in an effort to decrease all-cause injury incidence.

Concussion incidence increased from the 2014 to 2015 season, with no subsequent change in the following season. Concussion education and awareness has been increasing among players, policy makers, and clinicians potentially contributing to an increased awareness and, subsequent, increased incidence. Given the greater emphasis on player safety and concussion, the NFL continues to work to improve concussion education and awareness at all levels through various programs and educational campaigns.[20] One such program involved the introduction of independent certified athletic trainer spotters, who, in each game, assist in the identification of injuries on the field. In 2015, the medical timeout was implemented, allowing spotters who witnessed a player requiring assistance to alert referees to call a timeout on the field.[20] In addition, in 2015, new ruling was implemented allowing any medical personnel to call a medical timeout if they observed any player appearing disoriented and/or concussed.[21] This timeout is not charged to either team, and a substitution is allowed to be made.[21] The increase in concussion incidence in the 2015 season may be partially attributed to these new rules, which endeavor to improve the detection of concussions because they occur during the game play. However, we do recognize that increase in player size, speed, and strength is likely an additional factor for increases in concussion incidence.[22,23]

Given the nature of this epidemiological study, it is not possible to make any casual inferences on the increase in concussion incidence we found between 2014 and 2015. Likely, this increase in incidence is multifactorial. Although we did not see a significant decrease in concussion incidence in the 2016 season, a trend to a lower number of concussions in 2016 compared to 2015 was observed. This may be a result of a new policy to reduce concussions involving fines and forfeit of draft picks for teams that do not follow the NFL game day policy.[20] Whether this policy change and greater enforcement amounts to significant decrease in concussions should be assessed in the future.

This study has several limitations. First, the assumption that all injuries occurred during games may have overestimated the injury burden. However, previous studies have demonstrated that most injuries occur during games and not practices.[15,16,24] We have also used team-reported injury data, which requires reliance on each team to accurately report the injuries sustained. The validity of these injury reports as a reflection of the true NFL injury incidence has not been documented. However, teams are required by the NFL to report credible, accurate, and specific injury information or otherwise risk fines, suspensions, or forfeiture of draft picks. Although this does not guarantee validity and reliability of data, it does aid in the assumption that the data are accurate; however, it cannot negate the susceptibility of study findings to information bias.

Moreover, this study is limited in the ability to differentiate new injuries from previous. New injuries were defined as injuries that had not presented on the previous weeks' injury report. New injuries were not included in rate calculation for

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TABLE 4. Change in Concussion and Knee Injury Rate by Position per Year

	2012-2013 Injuries, n (%)		2013-2014 Injuries, n (%)		2014-2015 Injuries, n (%)		2015-2016 Injuries, n (%)		2016-2017 Injuries, n (%)		2012-2017 Total, n (%)	
Position	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury
Wide receiver	27 (17.1)	58 (16.1)	18 (12.6)	48 (11.9)	16 (14.2)	36 (9.3)	24 (13.3)	51 (12.0)	17 (10.8)	33 (10.7)	102 (13.6)	226 (12.0)
Tight end	18 (11.4)	23 (6.4)	17 (11.9)	33 (8.2)	9 (8.0)	28 (7.2)	17 (9.4)	33 (7.8)	14 (8.9)	30 (9.8)	75 (10.0)	147 (7.8)
Defensive back	42 (26.6)	60 (16.6)	40 (27.9)	58 (14.4)	40 (35.4)	76 (19.6)	51 (28.3)	77 (18.1)	50 (31.6)	56 (18.2)	223 (29.7)	327 (17.4)
Running back	17 (10.8)	44 (12.2)	16 (11.1)	45 (11.2)	10 (8.8)	30 (7.8)	13 (7.2)	43 (10.1)	12 (7.6)	26 (8.5)	68 (9.0)	188 (10.0)
Linebacker	11 (7.0)	44 (12.2)	13 (9.1)	59 (14.6)	12 (10.6)	74 (19.1)	20 (11.1)	60 (14.1)	18 (11.4)	52 (16.9)	74 (9.8)	289 (15.4)
Defensive line	13 (8.2)	56 (15.5)	13 (9.1)	64 (15.9)	5 (4.4)	71 (18.3)	19 (10.6)	53 (12.4)	10 (6.3)	43 (14.0)	60 (8.0)	287 (15.2)
Offensive line	23 (14.6)	61 (16.9)	20 (14.0)	78 (19.4)	19 (16.8)	63 (16.2)	26 (14.4)	88 (20.7)	28 (17.7)	60 (19.5)	116 (15.4)	350 (18.6)
Quarterback	7 (4.4)	6 (1.7)	6 (4.2)	14 (3.4)	2 (1.8)	6 (1.6)	10 (5.6)	9 (2.1)	8 (5.1)	7 (2.3)	33 (4.4)	42 (2.2)
Kicker, punter	0	9 (2.5)	0	3 (0.7)	0	3 (0.8)	0	11 (2.6)	1 (0.6)	0 (0)	1 (0.1)	26 (1.4)
Total	158 (100.0)	361 (100.0)	143 (100.0)	402 (100.0)	113 (100.0)	387 (100.0)	180 (100.0)	425 (100.0)	158 (100.0)	307 (100.0)	752 (100.0)	1882 (100.0)

	2012-2013 Injuries				2013-2014 Injuries				2014-2015 Injuries			
	Per 100 TG (95% CI)		Per 1000 AAR (95% CI)		Per 100 TG (95% CI)		Per 1000 AAR (95% CI)		Per 100 TG (95% CI)		Per 1000 AAR (95% CI)	
Position	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury
Wide receiver	5.49 (3.42-7.56)	11.79 (8.75-14.82)	4.99 (3.11-6.87)	10.72 (7.96-13.48)	3.66 (1.97-5.35)	9.70 (7.00 to 12.52)	3.33 (1.79-4.86)	8.87 (6.36 to 11.38)	3.25 (1.66 to 4.85)	7.32 (4.93 to 9.71)	2.96 (1.51 to 44.1)	6.65 (4.48 to 8.82)
Tight end	3.66 (1.97-5.35)	4.67 (2.76-6.59)	3.33 (1.79-4.86)	4.25 (2.51-5.99)	3.46 (1.81-5.10)	6.71 (4.42 to 9.00)	3.14 (1.65-4.63)	6.10 (4.02 to 8.18)	1.83 (0.63 to 3.02)	5.69 (3.58 to 7.80)	1.66 (0.58 to 2.75)	5.17 (3.26 to 7.09)
Defensive back	8.54 (5.95-11.12)	12.20 (9.11-15.28)	7.76 (5.41-10.11)	11.09 (8.28-13.89)	8.13 (5.61-10.65)	11.79 (8.75 to 14.82)	7.39 (5.10-9.68)	10.72 (7.96 to 13.48)	8.13 (5.61 to 10.65)	15.45 (11.97 to 18.92)	7.39 (5.10 to 9.68)	14.04 (10.89 to 17.20)
Running back	3.46 (1.81-5.10)	8.94 (6.30-11.59)	3.14 (1.65-4.63)	8.13 (5.73-10.53)	3.25 (1.66-4.85)	9.15 (6.47 to 11.82)	2.96 (1.51-44.1)	8.31 (5.89 to 10.74)	2.03 (0.77 to 3.29)	6.10 (3.92 to 8.28)	1.85 (0.70 to 2.99)	5.54 (3.56 to 7.53)
Linebacker	2.24 (0.91-3.56)	8.94 (6.30-11.59)	2.03 (0.83-3.23)	8.13 (5.73-10.53)	2.64 (1.21-4.08)	11.99 (8.93 to 15.05)	2.40 (1.10-3.71)	10.90 (8.12 to 13.68)	2.44 (1.06 to 3.82)	15.04 (11.61 to 18.47)	2.22 (0.96 to 3.47)	13.67 (10.56 to 16.79)
Defensive line	2.64 (1.21-4.08)	11.38 (8.40-14.36)	2.40 (1.10-3.71)	10.35 (7.64-13.06)	2.64 (1.21-4.08)	13.01 (9.82 to 16.20)	2.40 (1.10-3.71)	11.83 (8.93 to 14.72)	1.02 (0.13 to 1.91)	14.43 (11.07 to 17.79)	0.92 (0.11 to 1.73)	13.12 (10.07 to 16.17)
Offensive line	4.67 (2.76-6.59)	12.40 (9.29-15.51)	4.25 (2.51-5.99)	11.27 (8.44-14.10)	4.07 (2.28-5.85)	15.85 (12.34 to 19.37)	3.70 (2.08-5.32)	14.41 (11.21 to 17.61)	3.86 (2.13 to 5.60)	12.80 (9.64 to 15.97)	3.51 (1.93 to 5.09)	11.64 (8.77 to 14.52)
Quarterback	1.42 (0.37-2.48)	1.22 (0.24-2.20)	1.29 (0.34-2.25)	1.11 (0.22-2.00)	1.22 (0.24-2.20)	2.85 (1.35 to 4.34)	1.11 (0.22-2.00)	2.59 (1.23 to 3.94)	0.41 (−0.16 to 0.97)	1.22 (0.24 to 2.20)	0.37 (−0.14 to 0.88)	1.11 (0.22 to 2.00)
Kicker, punter	0	1.83 (0.63-3.02)	0	1.66 (0.58-2.75)	0	0.61 (−0.08 to 1.30)	0	0.55 (−0.07 to 1.18)	0	0.61 (−0.08 to 1.30)	0	0.55 (−0.07 to 1.18)
Total	32.11 (27.11-37.12)	73.37 (65.80-80.94)	29.19 (24.64-33.75)	66.70 (59.82-73.58)	29.07 (24.30-33.83)	81.71 (73.72 to 89.69)	26.42 (22.09-30.75)	74.28 (67.02 to 81.54)	22.97 (18.73 to 27.20)	78.66 (70.82 to 86.50)	20.88 (17.03 to 24.73)	71.51 (64.38 to 78.63)

	2015-2016 Injuries				2016-2017 Injuries				2012-2017 Total Injuries			
	Per 100 TG (95% CI)		Per 1000 AAR (95% CI)		Per 100 TG (95% CI)		Per 1000 AAR (95% CI)		Per 100 TG (95% CI)		Per 1000 AAR (95% CI)	
Position	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury
Wide receiver	4.88 (2.93-6.83)	10.37 (7.52-13.21)	4.43 (2.66-6.21)	9.42 (6.84-12.01)	3.46 (1.81 to 5.10)	6.71 (4.42-9.00)	3.14 (1.65 to 4.63)	6.10 (4.02-8.18)	20.73 (16.71 to 24.76)	9.19 (7.99-10.38)	18.85 (15.19 to 22.50)	41.76 (36.31-47.20)
Tight end	3.46 (1.81-5.10)	6.71 (4.42-9.00)	3.14 (1.65-4.63)	6.10 (4.02-8.18)	2.85 (1.35 to 4.34)	6.10 (3.92-8.28)	2.59 (1.23 to 3.94)	5.54 (3.56-7.53)	15.24 (11.79 to 18.69)	5.98 (5.01-6.94)	13.86 (10.72 to 16.99)	27.16 (22.77-31.55)
Defensive back	10.37 (7.52-13.21)	15.65 (12.15-19.15)	9.42 (6.84-12.01)	14.23 (11.05-17.41)	10.16 (7.35 to 12.98)	11.38 (8.40-14.36)	9.24 (6.68 to 11.80)	10.35 (7.64-13.06)	45.33 (39.38 to 51.27)	13.29 (11.85-14.73)	41.20 (35.80 to 46.61)	60.42 (53.87-66.97)
Running back	2.64 (1.21-4.08)	8.74 (6.13-11.35)	2.40 (1.10-3.71)	7.95 (5.57-10.32)	2.44 (1.06 to 3.82)	5.28 (3.25-7.32)	2.22 (0.96 to 3.47)	4.80 (2.96-6.65)	13.82 (10.54 to 17.11)	7.64 (6.55-8.73)	12.56 (9.58 to 15.55)	34.74 (29.77-39.70)
Linebacker	4.07 (2.28-5.85)	12.20 (9.11-15.28)	3.70 (2.08-5.32)	11.09 (8.28-13.89)	3.66 (1.97 to 5.35)	10.57 (7.70-13.44)	3.33 (1.79 to 4.86)	9.61 (7.00-12.22)	15.04 (11.61 to 18.47)	11.75 (10.39-13.10)	13.67 (10.56 to 16.79)	53.40 (47.24-59.56)

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TABLE 4. Change in Concussion and Knee Injury Rate by Position per Year *(Continued)*

	2015-2016 Injuries				2016-2017 Injuries				2012-2017 Total Injuries			
	Per 100 TG (95% CI)		Per 1000 AAR (95% CI)		Per 100 TG (95% CI)		Per 1000 AAR (95% CI)		Per 100 TG (95% CI)		Per 1000 AAR (95% CI)	
Position	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury	Concussion	Knee injury
Defensive line	3.86 (2.13-5.60)	10.77 (7.87-13.67)	3.51 (1.93-5.09)	9.79 (7.16-12.43)	3.66 (1.97 to 5.35)	8.74 (6.13-11.35)	3.33 (1.79 to 4.86)	7.95 (5.57-10.32)	12.20 (9.11 to 15.28)	11.67 (10.32-13.02)	11.09 (8.28 to 13.89)	53.03 (46.89-59.17)
Offensive line	5.28 (3.25-7.32)	17.89 (14.15-21.62)	4.80 (2.96-6.65)	16.26 (12.86-19.66)	5.69 (3.58 to 7.80)	12.20 (9.11-15.28)	5.17 (3.26 to 7.09)	11.09 (8.28-13.89)	23.58 (19.29 to 27.87)	14.23 (12.74-15.72)	21.43 (17.53 to 25.33)	64.67 (57.90-71.45)
Quarterback	3.66 (1.97-5.35)	1.83 (0.63-3.02)	3.33 (1.79-4.86)	1.66 (0.58-2.75)	1.63 (0.50 to 2.75)	1.42 (0.37-2.48)	1.48 (0.45 to 2.50)	1.29 (0.34-2.25)	6.71 (4.42 to 9.00)	1.71 (1.19-2.22)	6.10 (4.02 to 8.18)	7.76 (5.41-10.11)
Kicker, punter	0	2.24 (0.91-3.56)	0	2.03 (0.83-3.23)	0.20 (−0.20 to 0.60)	0	0.18 (−0.18 to 0.55)	0	0.20 (−0.20 to 0.60)	1.06 (0.65-1.46)	0.18 (−0.18 to 0.55)	4.80 (2.96-6.65)
Total	36.59 (31.24-41.93)	86.38 (78.17-94.59)	33.26 (28.40-38.12)	78.53 (71.06-86.00)	32.11 (27.11 to 37.12)	62.40 (55.42-69.38)	29.19 (24.64 to 33.75)	56.73 (50.38-63.07)	153.85 (141.92 to 163.77)	76.50 (73.05-79.96)	138.95 (129.02 to 148.88)	347.75 (332.03-363.46)

the subsequent concurrent reported weeks. However, the differentiation between novel, acute-on-chronic, and the reaggravation of previous injuries was unfeasible. Owing to the epidemiological nature of this study, there is also the possibility of selection bias. Although this study does have several limitations, the main goal was to compare injury incidence across the 5 seasons, and data were collected using the same methods and assumptions for all seasons; therefore, these assumptions should not make a significant difference in the between-year rate estimates of this study.

CONCLUSIONS

This study documented the incidence of NFL injuries over 5 seasons from 2012 to 2016 and observed an all-cause and concussion injury rates higher than that of other collision and contact sports. There was a significant increase in concussion incidence in the 2015 season compared with 2014. Comparatively, knee injury rates have significantly decreased in the 2016 season compared with 2014 and 2015. Consistently defensive backs incur the highest injury rates of all player positions, and future rule changes should be targeted to decrease injury among these players specifically. Concussion education and rule changes that favor decreasing concussion rates should continue to be implemented in football at all levels of play. The NFL should continue to develop strategies to decrease concussion rates and prioritize player safety.

References

1. Kerr ZY, Lynall RC, Mauntel TC, et al. High school football injury rates and services by athletic trainer employment status. *J Athl Train*. 2016;51:70–73.
2. Powell JW, Dompier TP. Analysis of injury rates and treatment patterns for time-loss and non-time-loss injuries among collegiate student-athletes. *J Athl Train*. 2004;39:56–70.
3. Lawrence DW, Hutchison MG, Comper P. Descriptive epidemiology of musculoskeletal injuries and concussions in the National Football League, 2012-2014. *Orthop J Sports Med*. 2015;3:2325967115583653.
4. Mullally WJ. Concussion. *Am J Med*. 2017;130:885–892.
5. Castellani RJ. Chronic traumatic encephalopathy: a paradigm in search of evidence? *Lab Invest*. 2015;95:576–584.
6. Golightly YM, Marshall SW, Callahan LF, et al. Early-onset arthritis in retired National Football League players. *J Phys Act Health* 2009;6:638–643.
7. Goldberg DS. Mild traumatic brain injury, the National Football League, and the manufacture of doubt: an ethical, legal, and historical analysis. *J Leg Med*. 2013;34:157–191.
8. Matava JM, Gortz S. The university of the National Football League: how technology, injury suveillance, and health care have improved the safety of America's game. *J Knee Surg*. 2016;29:370–378.
9. National Football League. 2016 rules changes and points of emphasis; 2017. Available at: http://operations.nfl.com/the-rules/2016-rules-changes-and-points-of-emphasis/. Accessed August 25, 2017.
10. Alosco ML, Jarnagin J, Tripodis Y, et al. Utility of providing a concussion definition in the assessment of concussion history in former NFL players. *Brain Inj*. 2017;31:1116–1123.
11. Casson IR, Viano DC, Powell JW, et al. Twelve years of National Football League concussion data. *Sports Health*. 2010;2:471–483.
12. Clark MD, Asken BM, Marshall SW, et al. Descriptive characteristics of concussions in National Football League games, 2010-2011 to 2013-2014. *Am J Sports Med*. 2017;45:929–936.
13. National Football League. Injuries; 2017. NFL Injuries Web site. Available at: http://www.nfl.com/injuries. Accessed July 11, 2017.
14. National Football League. 2016 NFL injury report policy; 2016. NFL Football Operations Web site. Available at: http://operations.nfl.com/updates/football-ops/2016-nfl-injury-report-policy/. Accessed July12, 2017.
15. Mall NA, Carlisle JC, Matava MJ, et al. Upper extremity injuries in the National Football League. Part I: hand and digital injuries. *Am J Sports Med*. 2008;36:1938–1944.
16. Lynch TS, Saltzman MD, Ghodasra JH, et al. Acromioclavicular joint injuries in the National Football League: epidemiology and management. *Am J Sports Med*. 2013;41:2904–2908.
17. McKay CD, Tufts RJ, Shaffer B, et al. The epidemiology of professional ice hockey injuries: a prospective report of six NHL seasons. *Br J Sports Med*. 2014;48:57–62.
18. Bathgate A, Best JP, Craig G, et al. A prospective study of injuries to elite Australian rugby union players. *Br J Sports Med*.. 2002;36:265–269; discussion 269.
19. The Associated Press. Players: crackdown on helmet hits could increase knee injuries; 2012. NFL News Web site. Available at: http://www.nfl.com/news/story/09000d5d81bb1c51/article/players-crackdown-on-helmet-hits-could-increase-knee-injuries. Published October 28, 2010. Accessed July 11, 2017.
20. National Football League. *2016 Health and Safety Report*. National Football League website. Available at: https://www.playsmartplaysafe.com/newsroom/reports/nfl-annual-player-health-and-safety-reports/. Accessed August 25, 2017.
21. Henne R. Six new rule changes adopted at NFL meetings; 2015. Chargers Web site. Available at: http://www.chargers.com/news/2015/03/24/six-new-rule-changes-adopted-nfl-meetings. Accessed September 4, 2017.
22. Kraemer WJ, Torine JC, Silvestre R, et al. Body size and composition of National Football League players. *J Strength Cond Res*. 2005;19:485–489.
23. Anzell AR, Potteiger JA, Kraemer WJ, et al. Changes in height, body weight, and body composition in American football players from 1942 to 2011. *J Strength Cond Res*. 2013;27:277–284.
24. Carlisle JC, Goldfarb CA, Mall N, et al. Upper extremity injuries in the National Football League. Part II: elbow, forearm, and wrist injuries. *Am J Sports Med*. 2008;36:1945–1952.

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